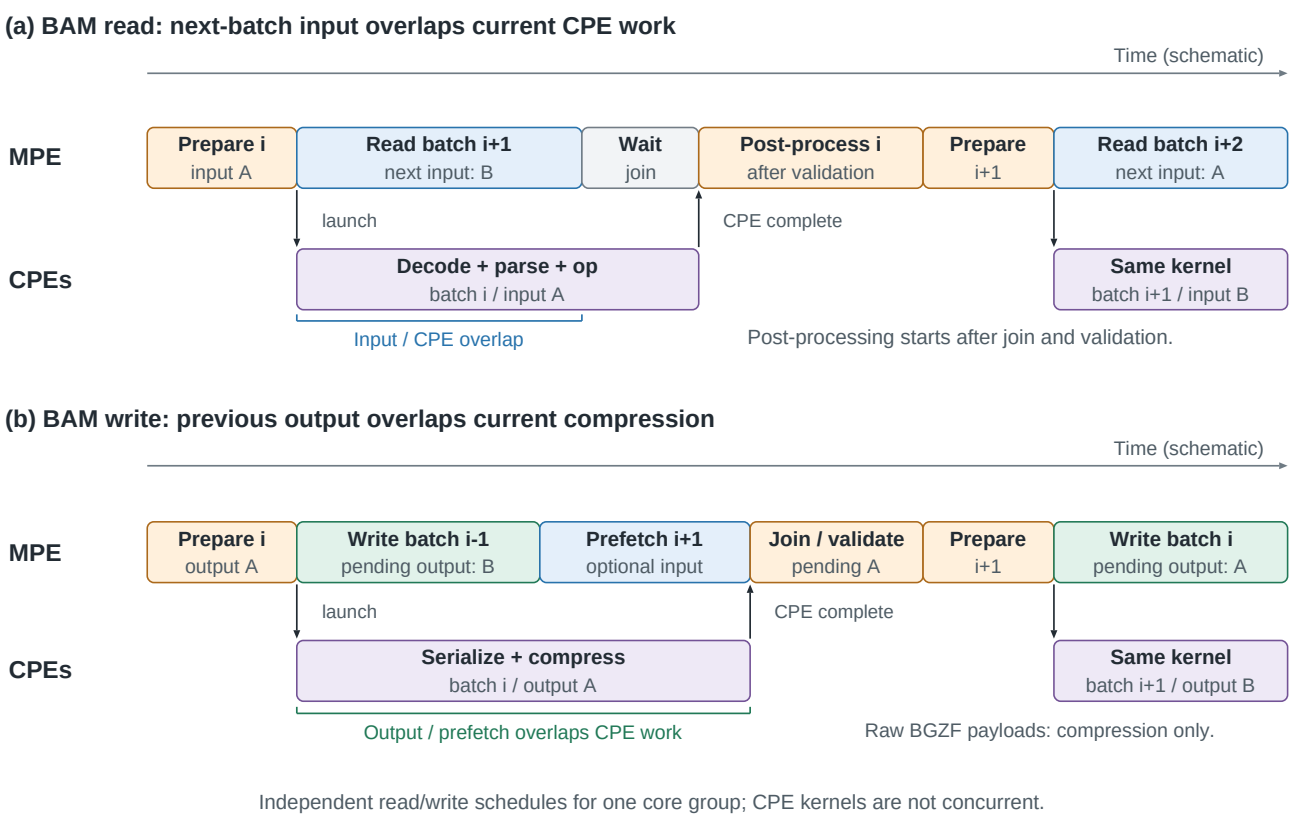
(a) BAM read: next-batch input overlaps current CPE work
Time (schematic)
MPE
Prepare i
input A
Read batch i+1
next input: B
Wait
join
Post-process i
after validation
Prepare
i+1
Read batch i+2
next input: A
launch
CPE complete
CPEs
Decode + parse + op
batch i / input A
Same kernel
batch i+1 / input B
Input / CPE overlap
Post-processing starts after join and validation.
(b) BAM write: previous output overlaps current compression
Time (schematic)
MPE
Prepare i
output A
Write batch i-1
pending output: B
Prefetch i+1
optional input
Join / validate
pending A
Prepare
i+1
Write batch i
pending output: A
launch
CPE complete
CPEs
Serialize + compress
batch i / output A
Same kernel
batch i+1 / output B
Output / prefetch overlaps CPE work
Raw BGZF payloads: compression only.
Independent read/write schedules for one core group; CPE kernels are not concurrent.